

Artwork: Test Image1013

Scale-Invariant Coordinate System

$$\tilde{X} = 2 \left(\frac{x}{W} \right) - 1, \quad \tilde{Y} = 1 - 2 \left(\frac{y}{H} \right) X = \frac{\left(\tilde{X} - \frac{5}{67} \right)}{\frac{83}{84}}, \quad Y = \frac{\left(\tilde{Y} + \frac{13}{176} \right)}{\frac{201}{182}}, \quad \theta = \arctan \left(\frac{Y}{X} \right)$$

Layer Definitions Boundary Manifolds

$$C_0(X, Y, v) = \frac{20 - 3v}{40} \cdot \left(1 + \frac{5}{133}X + \frac{7}{65}Y \right)$$

$$X_1 = \left(X - \frac{35}{159} \right), \quad Y_1 = \left(Y - \frac{99}{193} \right)$$

$$\tau_1(X, Y) = -\frac{7}{187} \cos^2 \left(\frac{351}{22}X_1 + \frac{365}{69}Y_1 - \frac{74}{141} \cos \left(\frac{581}{129}X_1 + \frac{370}{191}Y_1 \right) \right) + \frac{5}{76} \cos^4 \left(\frac{1077}{200}X_1 + \frac{2150}{107}Y_1 + \frac{93}{158} \cos \left(\frac{55}{31}X_1 + \frac{916}{169}Y_1 \right) \right) + \frac{1}{28} \cos^6 \left(\frac{1087}{39}X_1 + \frac{1891}{150}Y_1 + \frac{9}{15} \right)$$

$$X_2 = \left(X + \frac{33}{157} \right), \quad Y_2 = \left(Y + \frac{141}{194} \right)$$

$$\tau_2(X, Y) = \frac{18}{77} \cos^2 \left(\frac{1159}{71}X_2 + \frac{13}{2}Y_2 + \frac{166}{171} \cos \left(\frac{421}{184}X_2 + \frac{159}{61}Y_2 \right) \right) + \frac{13}{58} \cos^4 \left(\frac{1292}{197}X_2 + \frac{1447}{76}Y_2 + \frac{45}{172} \cos \left(\frac{334}{119}X_2 + \frac{802}{137}Y_2 \right) \right) + \frac{33}{128} \cos^6 \left(\frac{4477}{155}X_2 + \frac{2498}{195}Y_2 + \frac{20}{21} \right)$$

$$X_3 = \left(X + \frac{155}{179} \right), \quad Y_3 = \left(Y - \frac{29}{178} \right)$$

$$\tau_3(X, Y) = \frac{15}{184} \cos^2 \left(\frac{1769}{111}X_3 + \frac{371}{66}Y_3 + \frac{52}{103} \cos \left(\frac{85}{21}X_3 + \frac{246}{121}Y_3 \right) \right) - \frac{2}{173} \cos^4 \left(\frac{623}{100}X_3 + \frac{4036}{197}Y_3 - \frac{10}{19} \cos \left(\frac{169}{68}X_3 + \frac{767}{158}Y_3 \right) \right) - \frac{1}{40} \cos^6 \left(\frac{5305}{188}X_3 + \frac{1839}{155}Y_3 - \frac{59}{185} \right)$$

$$X_4 = \left(X - \frac{61}{178} \right), \quad Y_4 = \left(Y - \frac{56}{87} \right)$$

$$\tau_4(X, Y) = \frac{19}{176} \cos^2 \left(\frac{443}{28}X_4 + \frac{169}{28}Y_4 + \frac{9}{167} \cos \left(\frac{190}{51}X_4 + \frac{366}{169}Y_4 \right) \right) - \frac{9}{74} \cos^4 \left(\frac{1052}{185}X_4 + \frac{3739}{185}Y_4 - \frac{32}{97} \cos \left(\frac{144}{83}X_4 + \frac{958}{189}Y_4 \right) \right) - \frac{30}{197} \cos^6 \left(\frac{3528}{127}X_4 + \frac{2249}{186}Y_4 + \frac{23}{87} \right)$$

$$X_5 = \left(X + \frac{85}{139} \right), \quad Y_5 = \left(Y - \frac{140}{199} \right)$$

$$\tau_5(X, Y) = \frac{3}{136} \cos^4 \left(\frac{985}{169}X_5 + \frac{471}{23}Y_5 + \frac{39}{115} \cos \left(\frac{407}{170}X_5 + \frac{851}{149}Y_5 \right) \right) + \frac{2}{139} \cos^8 \left(\frac{1569}{134}X_5 + \frac{2279}{64}Y_5 - \frac{2}{7} \cos \left(\frac{310}{103}X_5 + \frac{1313}{189}Y_5 \right) \right) - \frac{1}{79} \cos^{16} \left(\frac{2461}{124}X_5 + \frac{7363}{131}Y_5 - \frac{2}{14} \right)$$

$$X_6 = \left(X - \frac{143}{194}\right), \quad Y_6 = \left(Y - \frac{82}{179}\right)$$

$$\tau_6(X, Y) = -\frac{2}{161} \cos^2 \left(\frac{3076}{197} X_6 + \frac{1051}{165} Y_6 - \frac{4}{37} \cos \left(\frac{107}{25} X_6 + \frac{57}{35} Y_6 \right) \right) - \frac{2}{191} \cos^4 \left(\frac{343}{60} X_6 + \frac{3975}{197} Y_6 + \frac{13}{188} \cos \left(\frac{484}{185} X_6 + \frac{523}{97} Y_6 \right) \right) - \frac{1}{73} \cos^6 \left(\frac{5021}{177} X_6 + \frac{1559}{132} Y_6 - \frac{1}{13} \cos \left(\frac{11}{46} X_6 + \frac{1198}{177} Y_6 \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_7 + \frac{4449}{124} Y_7 - \frac{1}{13} \cos \left(\frac{11}{46} X_7 + \frac{1198}{177} Y_7 \right) \right)$$

$$X_7 = \left(X + \frac{23}{127}\right), \quad Y_7 = \left(Y + \frac{65}{188}\right)$$

$$\tau_7(X, Y) = -\frac{1}{100} \cos^2 \left(\frac{1223}{87} X_7 + \frac{705}{97} Y_7 + \frac{154}{141} \cos \left(\frac{83}{14} X_7 + \frac{195}{199} Y_7 \right) \right) + \frac{1}{107} \cos^4 \left(\frac{1213}{177} X_7 + \frac{1595}{82} Y_7 - \frac{169}{152} \cos \left(\frac{11}{46} X_7 + \frac{1198}{177} Y_7 \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_7 + \frac{4449}{124} Y_7 - \frac{1}{13} \cos \left(\frac{11}{46} X_7 + \frac{1198}{177} Y_7 \right) \right)$$

$$X_8 = \left(X - \frac{99}{199}\right), \quad Y_8 = \left(Y + \frac{67}{140}\right)$$

$$\tau_8(X, Y) = \frac{47}{185} \cos^2 \left(\frac{1933}{122} X_8 + \frac{1041}{178} Y_8 - \frac{16}{105} \cos \left(\frac{687}{166} X_8 + \frac{421}{195} Y_8 \right) \right) + \frac{14}{65} \cos^4 \left(\frac{764}{131} X_8 + \frac{3531}{178} Y_8 + \frac{21}{130} \cos \left(\frac{329}{176} X_8 + \frac{907}{187} Y_8 \right) \right) - \frac{7}{71} \cos^6 \left(\frac{2868}{103} X_8 + \frac{939}{77} Y_8 + \frac{2}{13} \cos \left(\frac{11}{46} X_8 + \frac{1198}{177} Y_8 \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_8 + \frac{4449}{124} Y_8 - \frac{1}{13} \cos \left(\frac{11}{46} X_8 + \frac{1198}{177} Y_8 \right) \right)$$

$$X_9 = \left(X - \frac{28}{167}\right), \quad Y_9 = \left(Y - \frac{3}{10}\right)$$

$$\tau_9(X, Y) = -\frac{7}{130} \cos^2 \left(\frac{2282}{129} X_9 + \frac{773}{158} Y_9 + \frac{41}{44} \cos \left(\frac{270}{133} X_9 + \frac{318}{85} Y_9 \right) \right) - \frac{1}{94} \cos^4 \left(\frac{949}{177} X_9 + \frac{3590}{169} Y_9 - \frac{14}{23} \cos \left(\frac{99}{139} X_9 + \frac{1201}{200} Y_9 \right) \right) - \frac{3}{155} \cos^6 \left(\frac{2454}{199} X_9 + \frac{6179}{173} Y_9 + \frac{6}{13} \cos \left(\frac{11}{46} X_9 + \frac{1198}{177} Y_9 \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_9 + \frac{4449}{124} Y_9 - \frac{1}{13} \cos \left(\frac{11}{46} X_9 + \frac{1198}{177} Y_9 \right) \right)$$

$$X_{10} = \left(X - \frac{22}{43}\right), \quad Y_{10} = \left(Y + \frac{41}{86}\right)$$

$$\tau_{10}(X, Y) = -\frac{5}{186} \cos^2 \left(\frac{1498}{95} X_{10} + \frac{1239}{200} Y_{10} + \frac{8}{37} \cos \left(\frac{589}{141} X_{10} + \frac{332}{179} Y_{10} \right) \right) + \frac{3}{73} \cos^4 \left(\frac{710}{121} X_{10} + \frac{2605}{129} Y_{10} + \frac{21}{199} \cos \left(\frac{47}{22} X_{10} + \frac{140}{27} Y_{10} \right) \right) - \frac{22}{151} \cos^6 \left(\frac{3324}{119} X_{10} + \frac{16}{13} \cos \left(\frac{11}{46} X_{10} + \frac{1198}{177} Y_{10} \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_{10} + \frac{4449}{124} Y_{10} - \frac{1}{13} \cos \left(\frac{11}{46} X_{10} + \frac{1198}{177} Y_{10} \right) \right)$$

$$X_{11} = \left(X - \frac{101}{177}\right), \quad Y_{11} = \left(Y + \frac{105}{197}\right)$$

$$\tau_{11}(X, Y) = -\frac{33}{119} \cos^2 \left(\frac{2966}{183} X_{11} + \frac{1053}{184} Y_{11} + \frac{41}{166} \cos \left(\frac{893}{196} X_{11} + \frac{317}{198} Y_{11} \right) \right) - \frac{3}{173} \cos^4 \left(\frac{548}{95} X_{11} + \frac{3830}{187} Y_{11} - \frac{52}{199} \cos \left(\frac{167}{74} X_{11} + \frac{266}{55} Y_{11} \right) \right) - \frac{22}{195} \cos^6 \left(\frac{4309}{153} X_{11} + \frac{6}{13} \cos \left(\frac{11}{46} X_{11} + \frac{1198}{177} Y_{11} \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_{11} + \frac{4449}{124} Y_{11} - \frac{1}{13} \cos \left(\frac{11}{46} X_{11} + \frac{1198}{177} Y_{11} \right) \right)$$

$$X_{12} = \left(X - \frac{7}{12}\right), \quad Y_{12} = \left(Y + \frac{27}{64}\right)$$

$$\tau_{12}(X, Y) = -\frac{11}{83} \cos^2 \left(\frac{2095}{122} X_{12} + \frac{1046}{171} Y_{12} + \frac{129}{196} \cos \left(\frac{648}{145} X_{12} + \frac{295}{101} Y_{12} \right) \right) + \frac{7}{194} \cos^4 \left(\frac{616}{125} X_{12} + \frac{2964}{155} Y_{12} + \frac{59}{36} \cos \left(\frac{34}{23} X_{12} + \frac{273}{68} Y_{12} \right) \right) + \frac{14}{123} \cos^6 \left(\frac{4976}{175} X_{12} + \frac{22}{13} \cos \left(\frac{11}{46} X_{12} + \frac{1198}{177} Y_{12} \right) \right) + \frac{1}{91} \cos^8 \left(\frac{1814}{149} X_{12} + \frac{4449}{124} Y_{12} - \frac{1}{13} \cos \left(\frac{11}{46} X_{12} + \frac{1198}{177} Y_{12} \right) \right)$$

$$X_{13} = \left(X - \frac{48}{199}\right), \quad Y_{13} = \left(Y + \frac{20}{39}\right)$$

$$\tau_{13}(X, Y) = -\frac{23}{132} \cos^2 \left(\frac{2691}{163} X_{13} + \frac{417}{88} Y_{13} - \frac{77}{152} \cos \left(4X_{13} + \frac{548}{177} Y_{13} \right) \right) + \frac{7}{188} \cos^4 \left(\frac{997}{161} X_{13} + \frac{3845}{199} Y_{13} + \frac{11}{40} \cos \left(\frac{191}{64} X_{13} + \frac{1126}{179} Y_{13} \right) \right) + \frac{7}{57} \cos^6 \left(\frac{2365}{82} X_{13} + 10.0 \right)$$

$$X_{14} = \left(X - \frac{46}{81}\right), \quad Y_{14} = \left(Y + \frac{74}{187}\right)$$

$$\tau_{14}(X, Y) = -\frac{11}{140} \cos^2 \left(\frac{2690}{177} X_{14} + \frac{397}{97} Y_{14} - \frac{13}{187} \cos \left(\frac{596}{147} X_{14} + \frac{269}{161} Y_{14} \right) \right) - \frac{6}{199} \cos^4 \left(\frac{1134}{181} X_{14} + \frac{1026}{55} Y_{14} - \frac{105}{181} \cos \left(\frac{257}{131} X_{14} + \frac{83}{14} Y_{14} \right) \right) - \frac{2}{83} \cos^6 \left(\frac{3823}{141} X_{14} + \frac{2}{11} \right)$$

$$X_{15} = \left(X + \frac{11}{197}\right), \quad Y_{15} = \left(Y - \frac{70}{199}\right)$$

$$\tau_{15}(X, Y) = \frac{40}{139} \cos^2 \left(\frac{374}{27} X_{15} + \frac{1158}{155} Y_{15} - \frac{125}{117} \cos \left(\frac{387}{64} X_{15} + \frac{127}{42} Y_{15} \right) \right) - \frac{1}{84} \cos^4 \left(\frac{1031}{199} X_{15} + \frac{2272}{109} Y_{15} - \frac{23}{125} \cos \left(\frac{61}{59} X_{15} + \frac{962}{165} Y_{15} \right) \right) + \frac{16}{153} \cos^6 \left(\frac{1402}{53} X_{15} + \frac{13}{11} \right)$$

Master Composition (Recursive Layer Synthesis)

$$H_v^{(0)}(X, Y) = C_0(X, Y, v)$$

$$H_v^1(X, Y) = (1 - W_1(X, Y)) \cdot H_v^0(X, Y) + W_1(X, Y) \cdot C_1(X, Y, v)$$

$$H_v^2(X, Y) = (1 - W_2(X, Y)) \cdot H_v^1(X, Y) + W_2(X, Y) \cdot C_2(X, Y, v)$$

$$H_v^3(X, Y) = (1 - W_3(X, Y)) \cdot H_v^2(X, Y) + W_3(X, Y) \cdot C_3(X, Y, v)$$

$$H_v^4(X, Y) = (1 - W_4(X, Y)) \cdot H_v^3(X, Y) + W_4(X, Y) \cdot C_4(X, Y, v)$$

$$H_v^5(X, Y) = (1 - W_5(X, Y)) \cdot H_v^4(X, Y) + W_5(X, Y) \cdot C_5(X, Y, v)$$

$$H_v^6(X, Y) = (1 - W_6(X, Y)) \cdot H_v^5(X, Y) + W_6(X, Y) \cdot C_6(X, Y, v)$$

$$H_v^7(X, Y) = (1 - W_7(X, Y)) \cdot H_v^6(X, Y) + W_7(X, Y) \cdot C_7(X, Y, v)$$

$$H_v^8(X, Y) = (1 - W_8(X, Y)) \cdot H_v^7(X, Y) + W_8(X, Y) \cdot C_8(X, Y, v)$$

$$H_v^9(X, Y) = (1 - W_9(X, Y)) \cdot H_v^8(X, Y) + W_9(X, Y) \cdot C_9(X, Y, v)$$

$$H_v^{10}(X, Y) = (1 - W_{10}(X, Y)) \cdot H_v^9(X, Y) + W_{10}(X, Y) \cdot C_{10}(X, Y, v)$$

$$H_v^{11}(X, Y) = (1 - W_{11}(X, Y)) \cdot H_v^{10}(X, Y) + W_{11}(X, Y) \cdot C_{11}(X, Y, v)$$

$$H_v^{12}(X, Y) = (1 - W_{12}(X, Y)) \cdot H_v^{11}(X, Y) + W_{12}(X, Y) \cdot C_{12}(X, Y, v)$$

$$H_v^{13}(X, Y) = (1 - W_{13}(X, Y)) \cdot H_v^{12}(X, Y) + W_{13}(X, Y) \cdot C_{13}(X, Y, v)$$

$$H_v^{14}(X, Y) = (1 - W_{14}(X, Y)) \cdot H_v^{13}(X, Y) + W_{14}(X, Y) \cdot C_{14}(X, Y, v)$$

$$H_v^{15}(X, Y) = (1 - W_{15}(X, Y)) \cdot H_v^{14}(X, Y) + W_{15}(X, Y) \cdot C_{15}(X, Y, v)$$

$$H_v(X, Y) = H_v^{15}(X, Y)$$

Final RGB Channel Evaluation

$$\text{Color}(x, y) = (F(H_0(X, Y)), F(H_1(X, Y)), F(H_2(X, Y)))$$

where $F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$ performs exact differentiable color clamping to $[0, 255]$.